Mechanical Systems & Structural Design.

Robot structure overview

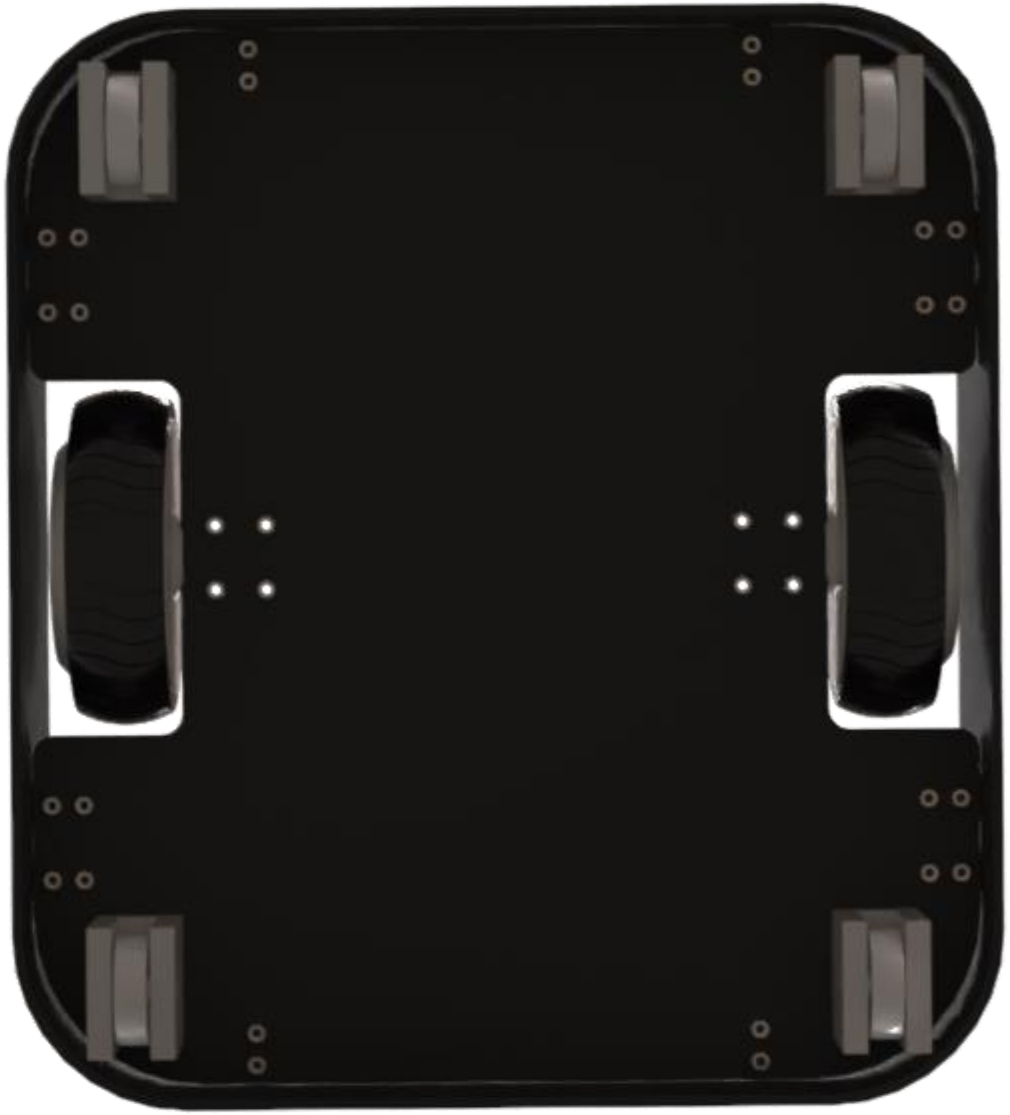
- Our robot consist of 4 main parts :
- Base
- Aluminum chassis
- Body
- Hands

Steel-Base

- **Material Choice:** We chose **5mm carbon steel sheets** for the construction. This thickness provides extreme durability and, more importantly, a **low Center of Gravity**. This makes the robot 'bottom-heavy,' which is essential for stability when the arms are extended up to 85 cm.
- **Modular Assembly:** As you can see in the exploded view, the base consists of three main parts. This design allows us to house the heavy hoverboard motors and batteries securely inside while maintaining easy access for assembly.
- **Precision Manufacturing:** By using custom-bent steel, we ensure that the internal aluminum chassis has a perfectly rigid mounting point, which eliminates vibrations during autonomous navigation."

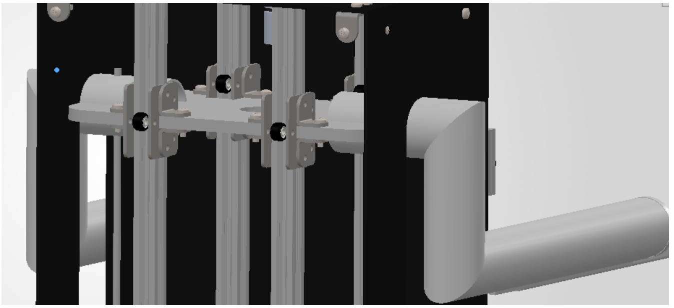
Wheels

- **Drive System:** Powered by two high-torque **hoverboard motors** capable of handling heavy payloads.
- **Stabilization:** Integrated with **4 industrial caster wheels** to ensure a perfectly level platform during movement.
- **6-Point Contact:** Strategic wheel placement provides maximum balance and smooth navigation over various indoor floor surfaces.
- **Maintenance:** Modular wheel assemblies allow for rapid replacement or servicing via a specialized 24-bolt pattern.



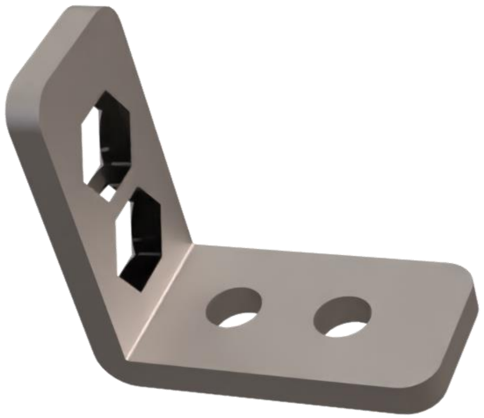
V-Slots aluminium profile chassis

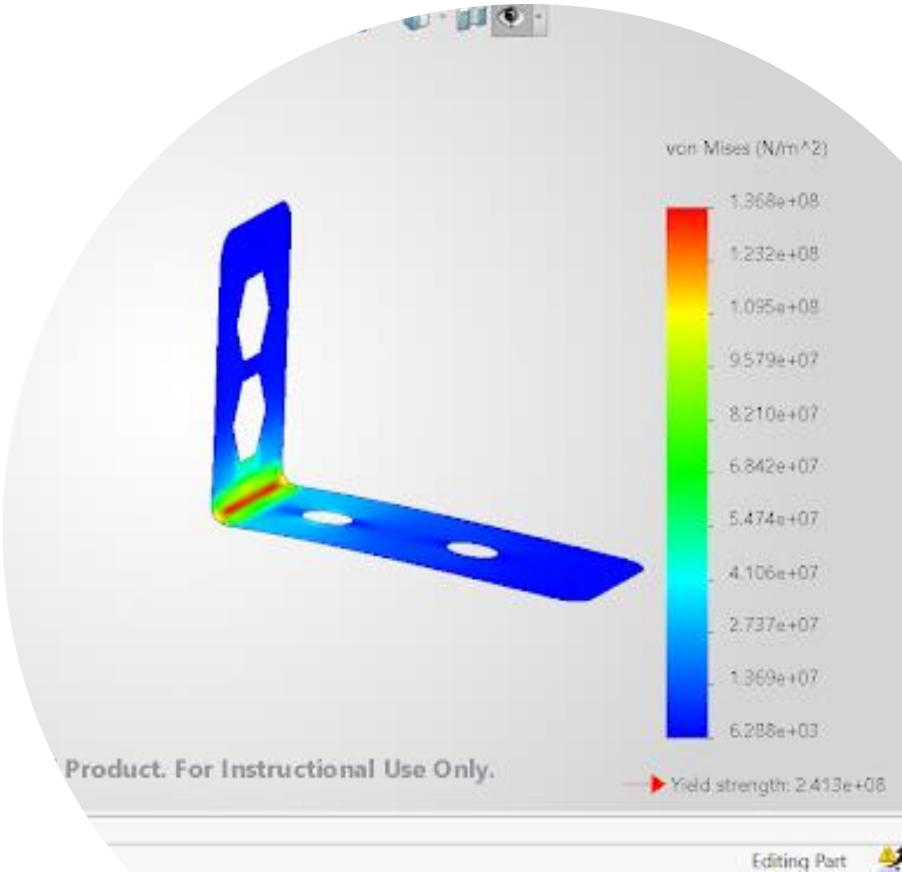
- **Internal Skeleton:** A modular frame that supports all internal boards and the vertical lifting mechanism.
- **Dual Purpose:** Profiles act as **structural spines** and **precision linear rails** for the bearing wheels of the arms.
- **Modularity:** Vertical design allows for easy height adjustment of electronics trays for optimal wire management



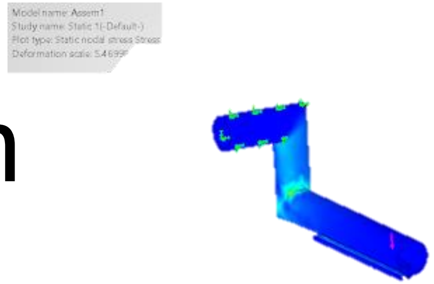
Customized L Bracket

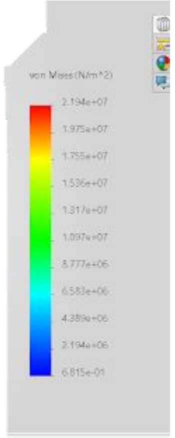
- **Integrated Nut-Locking:** Features unique **hexagonal cutouts** designed to trap and lock nuts in place during assembly.
- **Single-Tool Maintenance:** Eliminates the need for a second wrench inside the chassis, enabling rapid, one-handed servicing and assembly.
- **Validated Structural Integrity:** FEA results confirm a maximum stress of **136.8 MPa**, providing a safe operational margin against the **241.3 MPa yield strength**.
- **Optimized Load Transfer:** Specifically engineered to bridge the connection between the modular aluminum frame and the heavy-duty steel base.

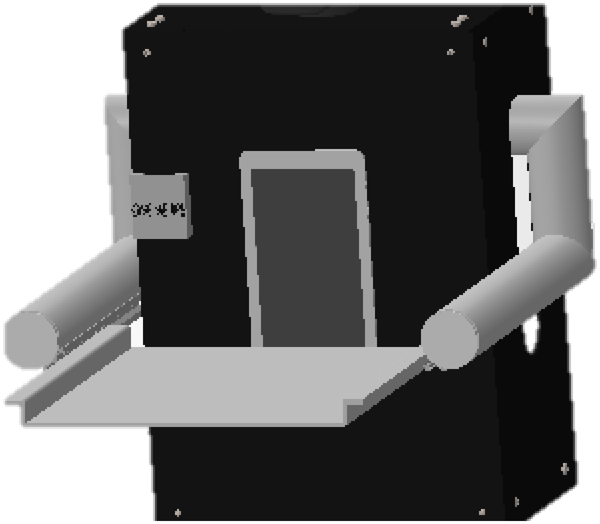




Hand Mechanism

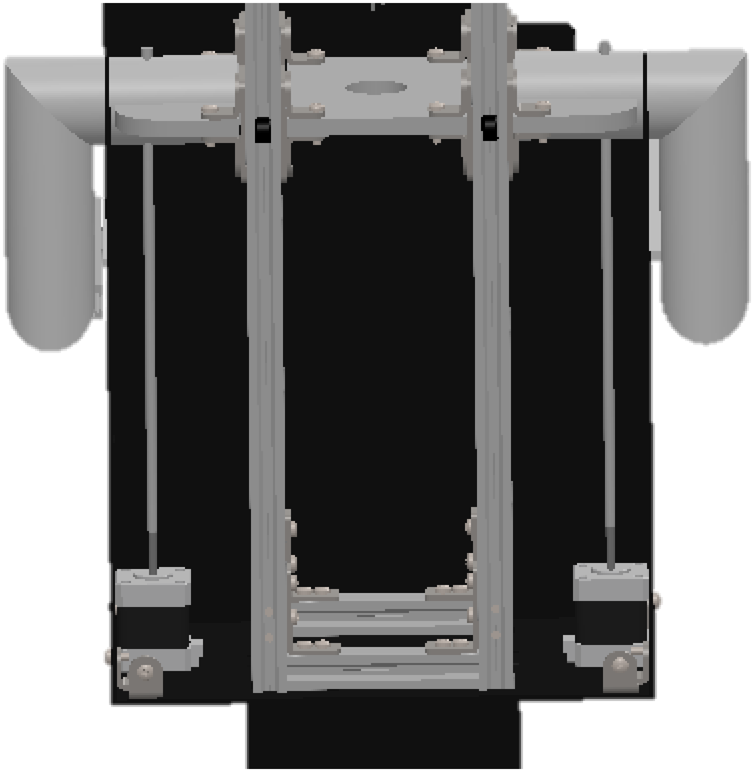






The Lifting Mechanism

- **Linear Transformation:** Your mechanism uses a **Lead Screw** or **Belt-Drive** (integrated with the V-slot) to convert the rotational torque of the NEMA 17 into linear vertical force.
- **Mechanical Advantage:** By using a lead screw (typically 8mm lead), the motor doesn't have to "fight" the full weight of the tray directly. One full rotation only moves the arm a few millimeters, which "multiplies" the effective lifting power.
- **Stability at Rest:** Stepper motors provide high **Holding Torque,** meaning the food tray stays at exactly 85 cm even when the power is on, without needing a mechanical brake, as it give about **0.4–0.6 Nm**.



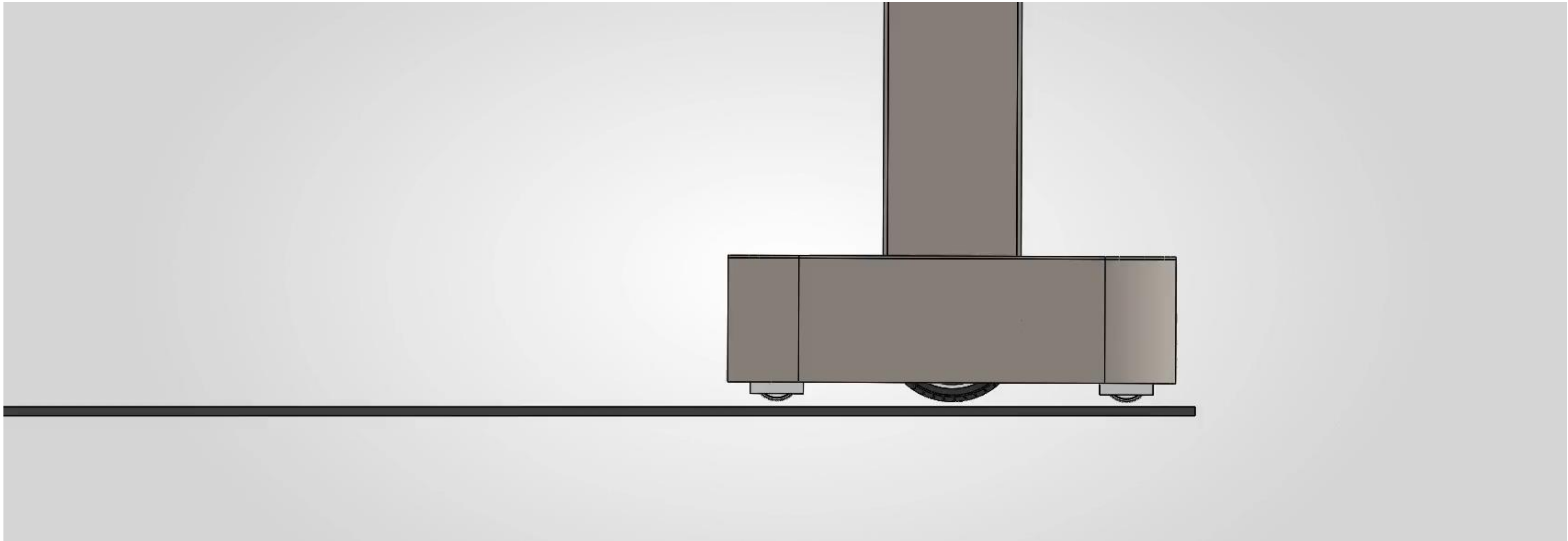
Head

- **Material:** 3D-printed **ABS Plastic** for a high strength-to-weight ratio and impact durability.
- **Active Pan Mechanism:** Integrated with a high-precision motor to provide autonomous rotation for interactive customer engagement and environmental scanning.
- **Sensory Housing:** Specifically engineered to contain the **Main Navigation Camera**, providing it with a protected, elevated vantage point.
- **Optimized Inertia:** Lightweight plastic construction reduces the torque requirements for the neck motor, ensuring smooth and rapid head movements.

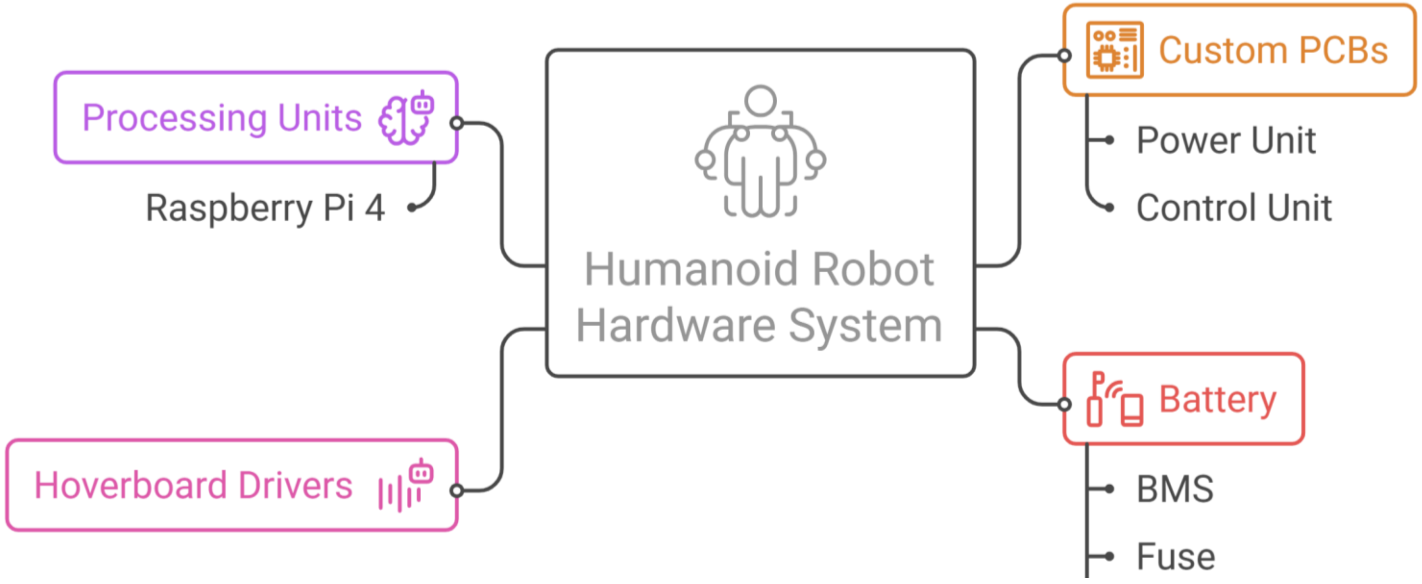
Frame

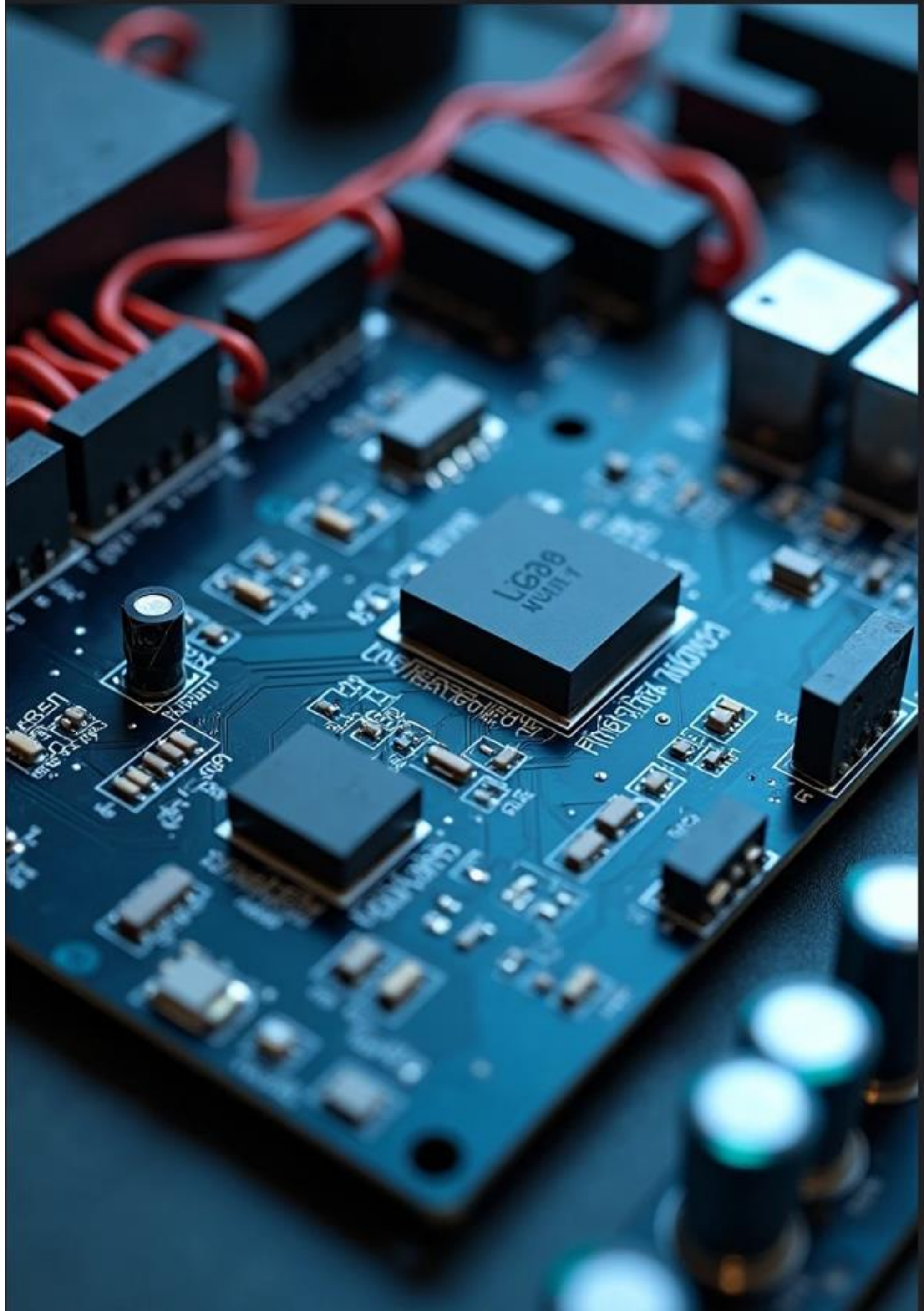
- **Material:** Constructed from **3mm Black Acrylic sheets** for a sleek, professional finish.
- **Lightweight Protection:** Provides a lightweight external "skin" that protects internal V-slot structures and electronic components from dust and interference.
- **Manufacturing Method:** Precision laser-cut panels designed for easy assembly and a seamless fit over the 5mm steel base.
- **Aesthetic Design:** The opaque black finish hides internal wiring and sensors, giving the robot a clean, modern "waiter" appearance suitable for hospitality environments.

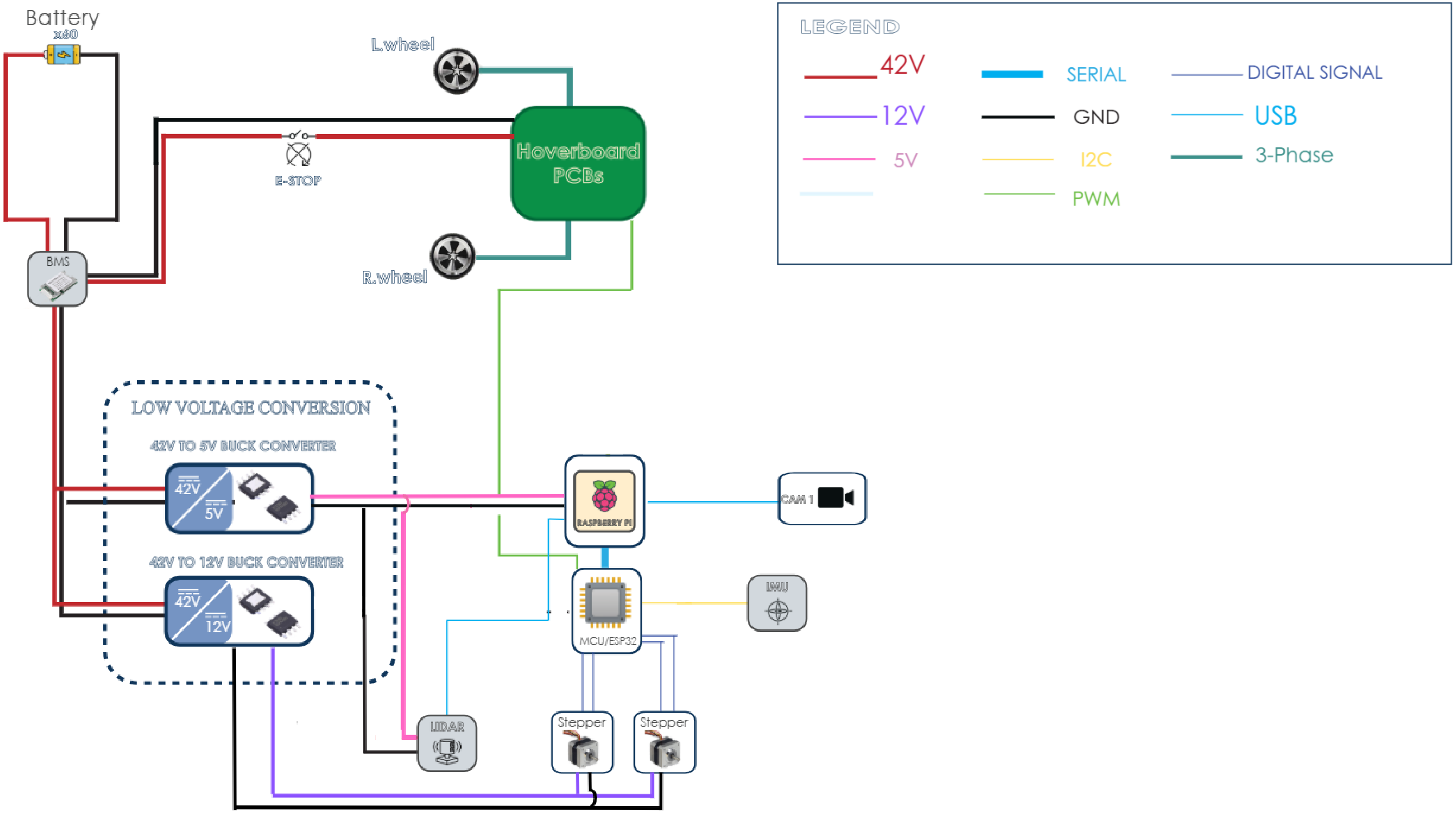
Animation



Hardware System







Battery
x60
L.wheel
R.wheel
E-STOP
Hoverboard PCBs
BMS
LEGEND
42V
12V
5V
SERIAL
GND
I2C
PWM
DIGITAL SIGNAL
USB
3-Phase
LOW VOLTAGE CONVERSION
42V TO 5V BUCK CONVERTER
42V
5V
42V TO 12V BUCK CONVERTER
42V
12V
RASPBERRY PI
CAM 1
MCU/ESP32
IMU
LIDAR
Stepper
Stepper

Hover Board PCB

BLDC Motor

A 3-phase Brushless DC motor is integrated into each wheel.

A single hoverboard motor driver board is used to control two hoverboard wheels simultaneously.

Hoverboard driver PCB

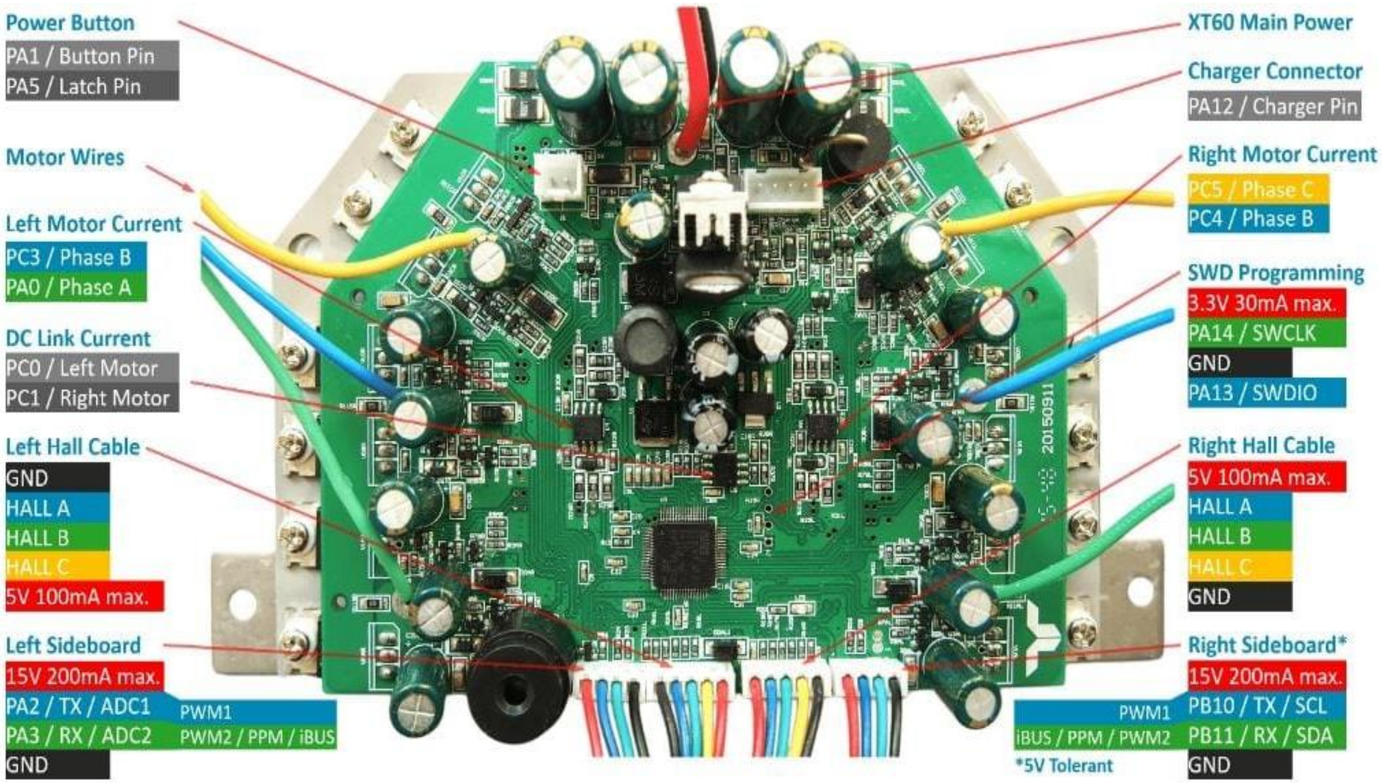
Hall/Encoder Feedback

Built-in Hall effect sensors and encoders provide speed and position estimation.

Receives PWM control signals from the main controller (ESP32)

PWM is used to command wheel speed and direction

Control Signal



Custom Power Distribution PCB

Power Distribution

Distributes battery power to various electronic components.

Electronics Protection

Protects sensitive electronics from voltage spikes and surges.

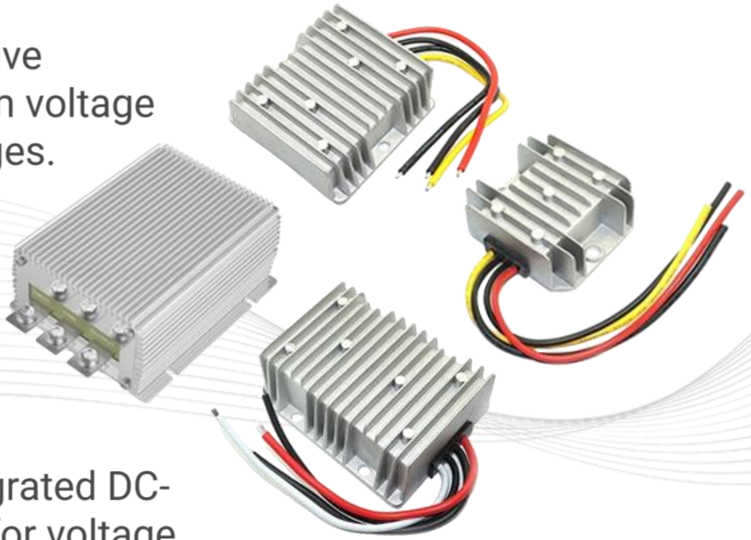
Voltage Generation

Generates required voltage rails for different electronic circuits.

Integrated Converters

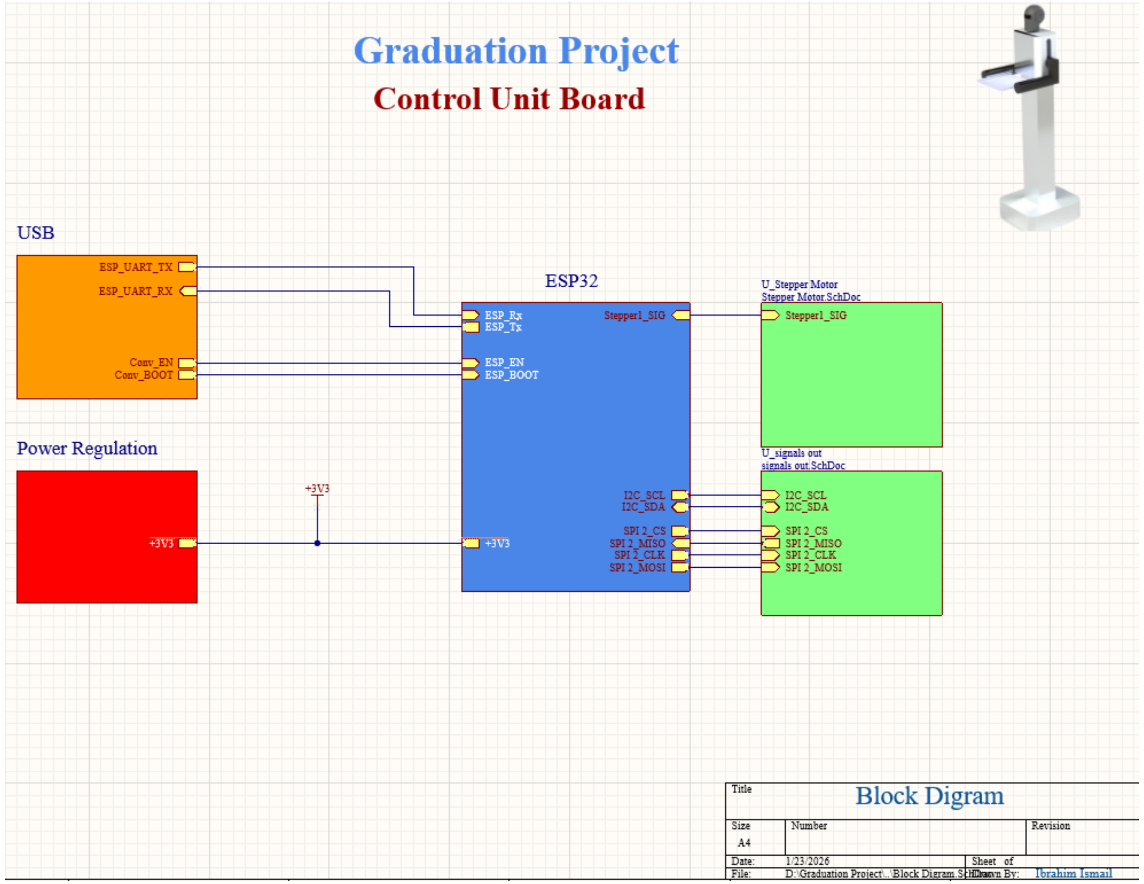
Includes 3 integrated DC-DC converters for voltage conversion.

- 42 V → 12 V
- 42 V → 5 V
- Spare DC-DC Converter

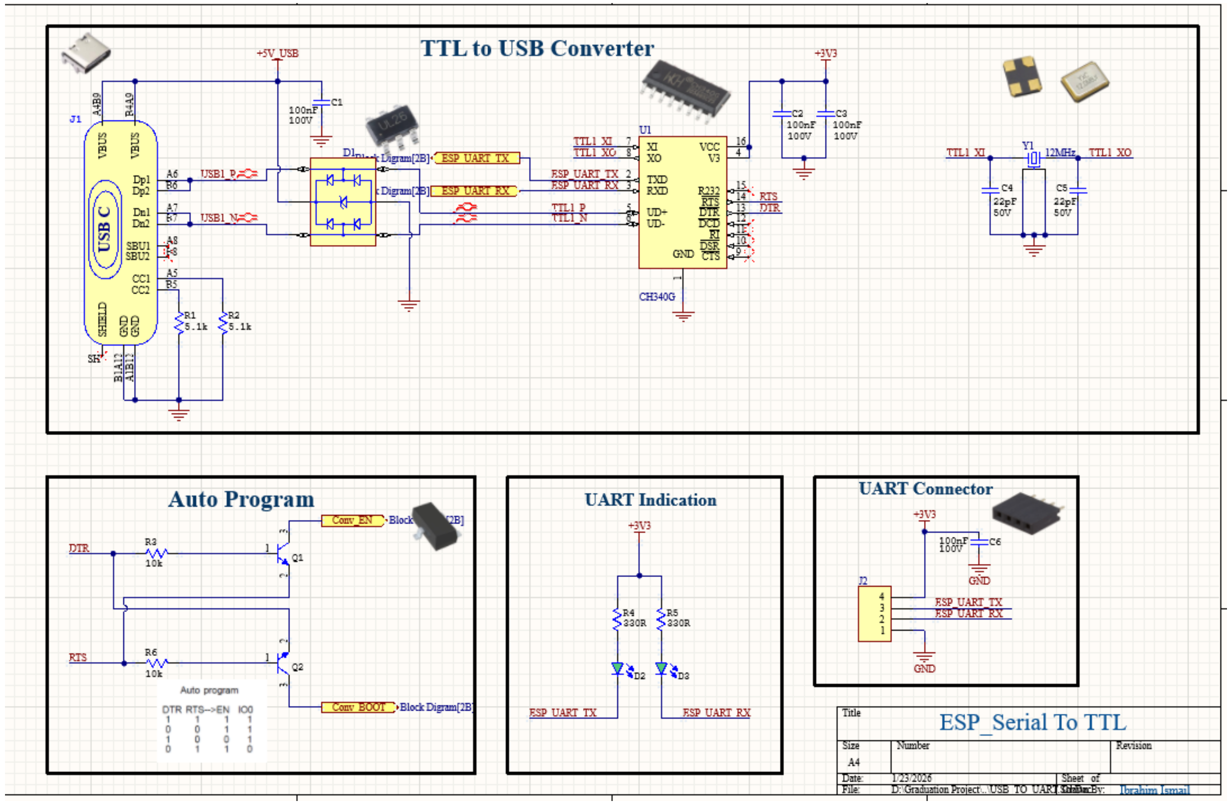


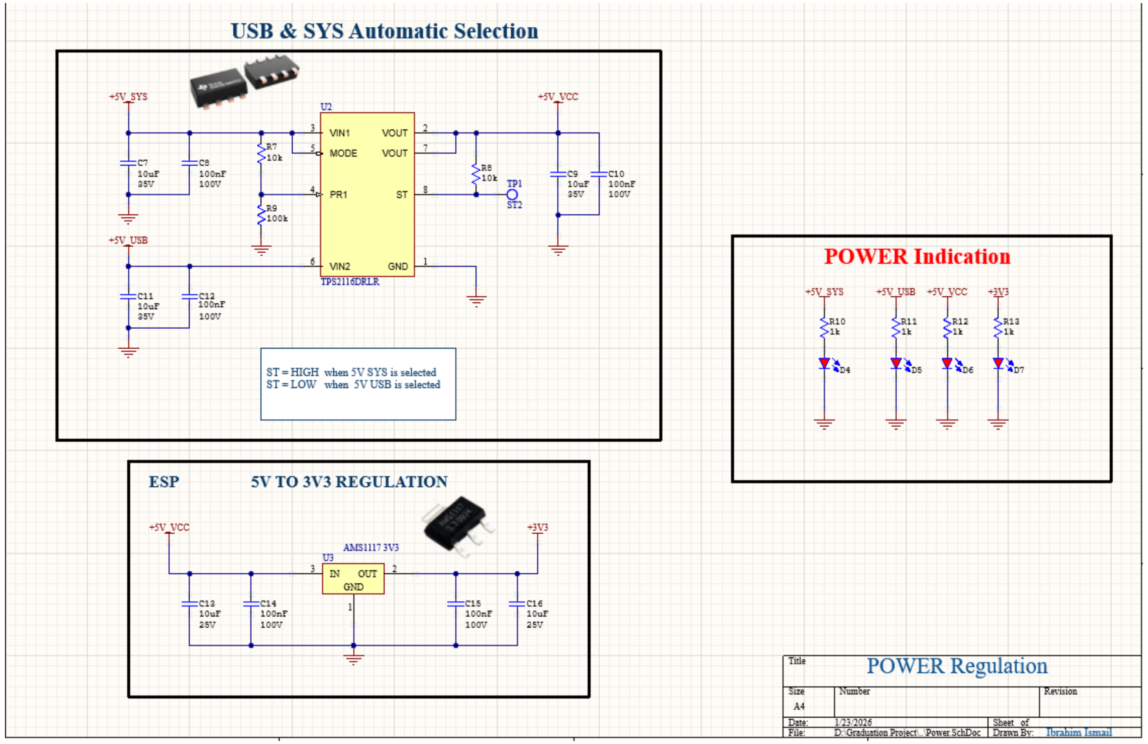
Custom Unit PCB

- The Control Unit Board **contains** the microcontroller (ESP32)
- **Responsible for** controlling the motion of the Robot and taking feedback from various sensors, such as IMU
- **Aesthetic Design:** The opaque black finish hides internal wiring and sensors, giving the robot a clean, modern "waiter" appearance suitable for hospitality environments.

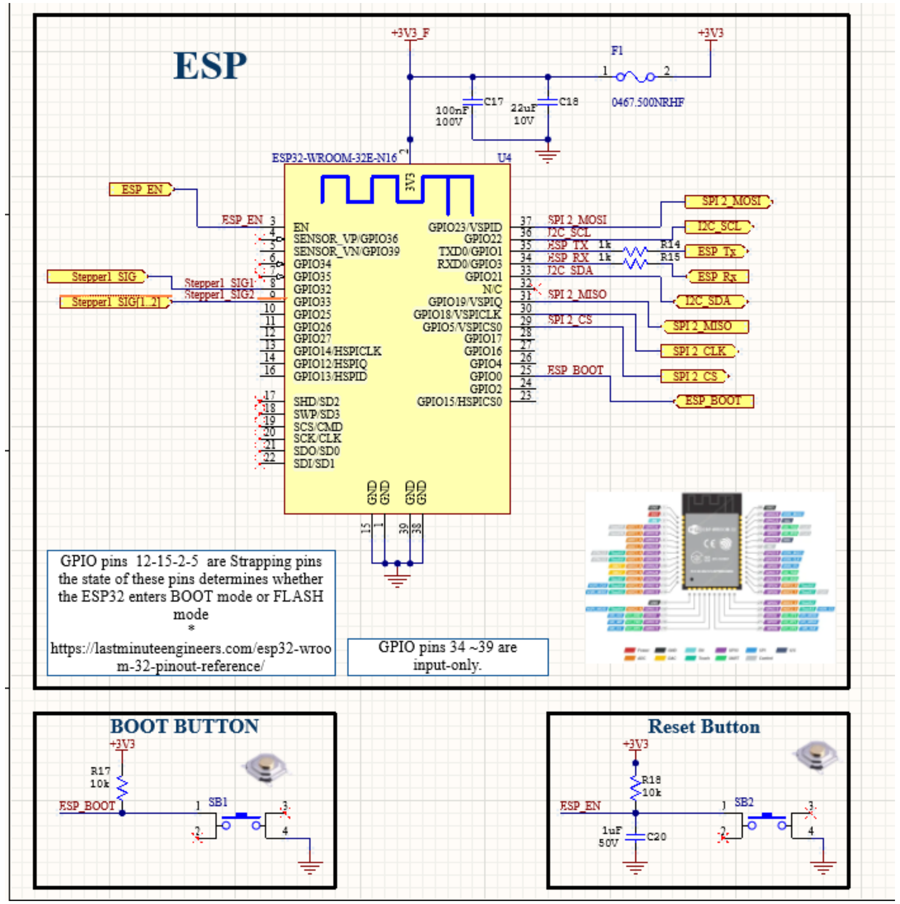


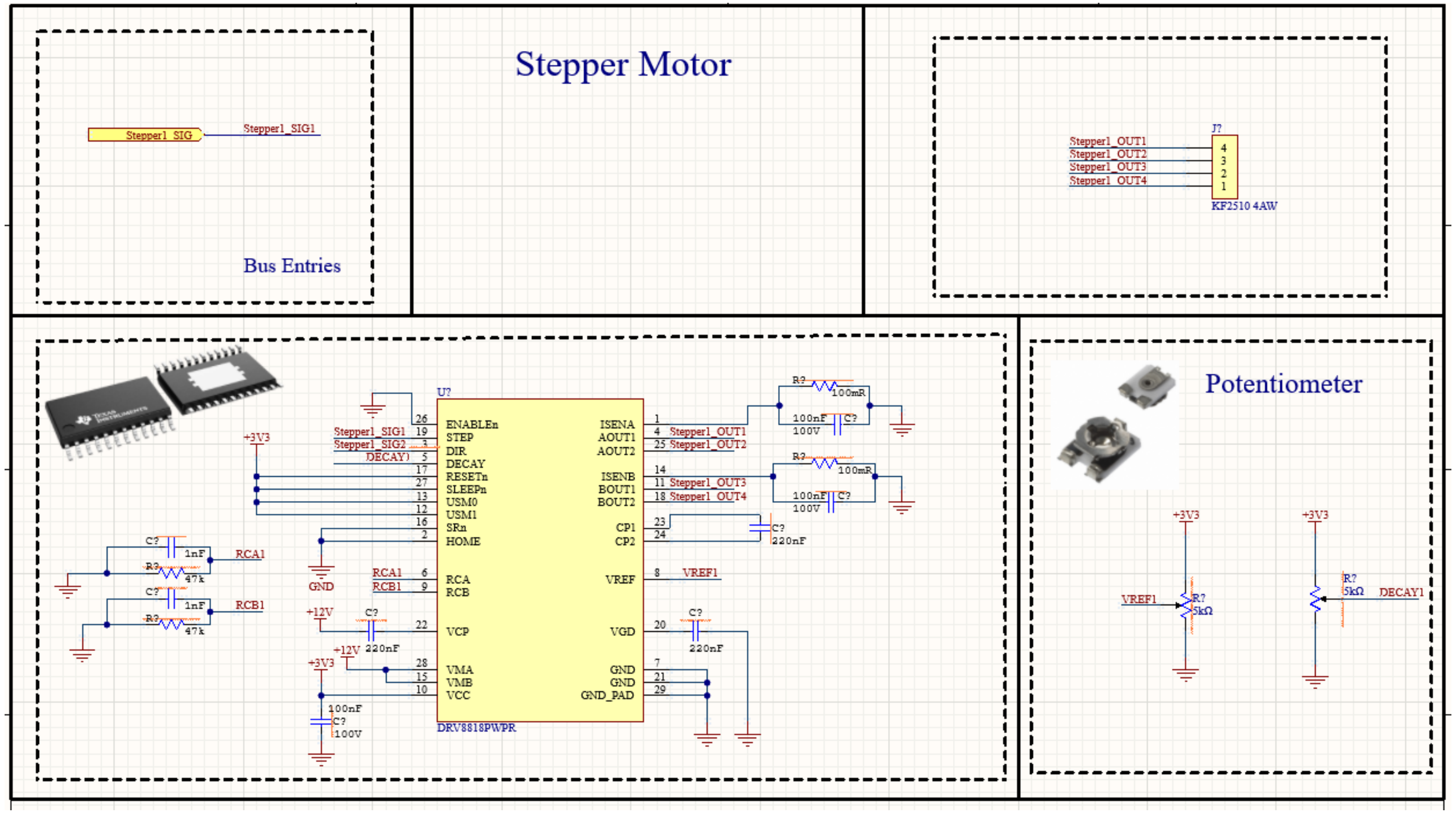
Schematics for PCB





Schematics for PCB



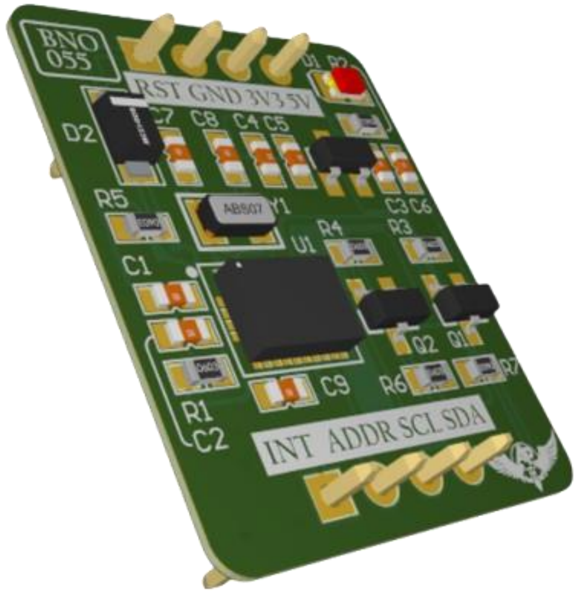


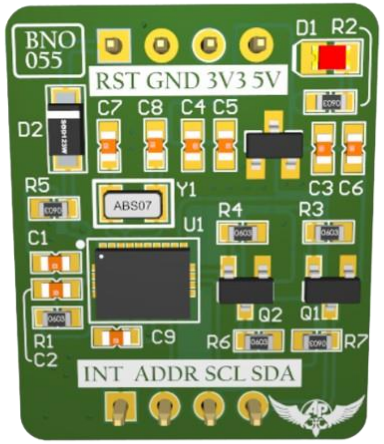
Perception Sensors: LiDAR & IMU (BNO055)

1- LiDAR Sensor
•Provides **360° environment scanning** for indoor navigation
•Used for:
- Obstacle detection
- Mapping and localization (SLAM)
- Safe navigation in dynamic restaurant environment

2- IMU Sensor (BNO055)
•9-DOF absolute orientation sensor:
- Accelerometer
- Gyroscope
- Magnetometer

•Mounted on the **control unit PCB**
•Used for:
- Orientation estimation
- Robot posture awareness
- Motion stability and smooth control





High-Level Computing – Raspberry Pi 4

❑ **Why Raspberry Pi is Used**

•Acts as the **main high-level processor** of the humanoid robot

•Handles computationally intensive tasks that are not suitable for microcontrollers

•Separates **real-time control** (ESP32) from **AI and perception processing**

❑ **Core Responsibilities**

•**Computer Vision**

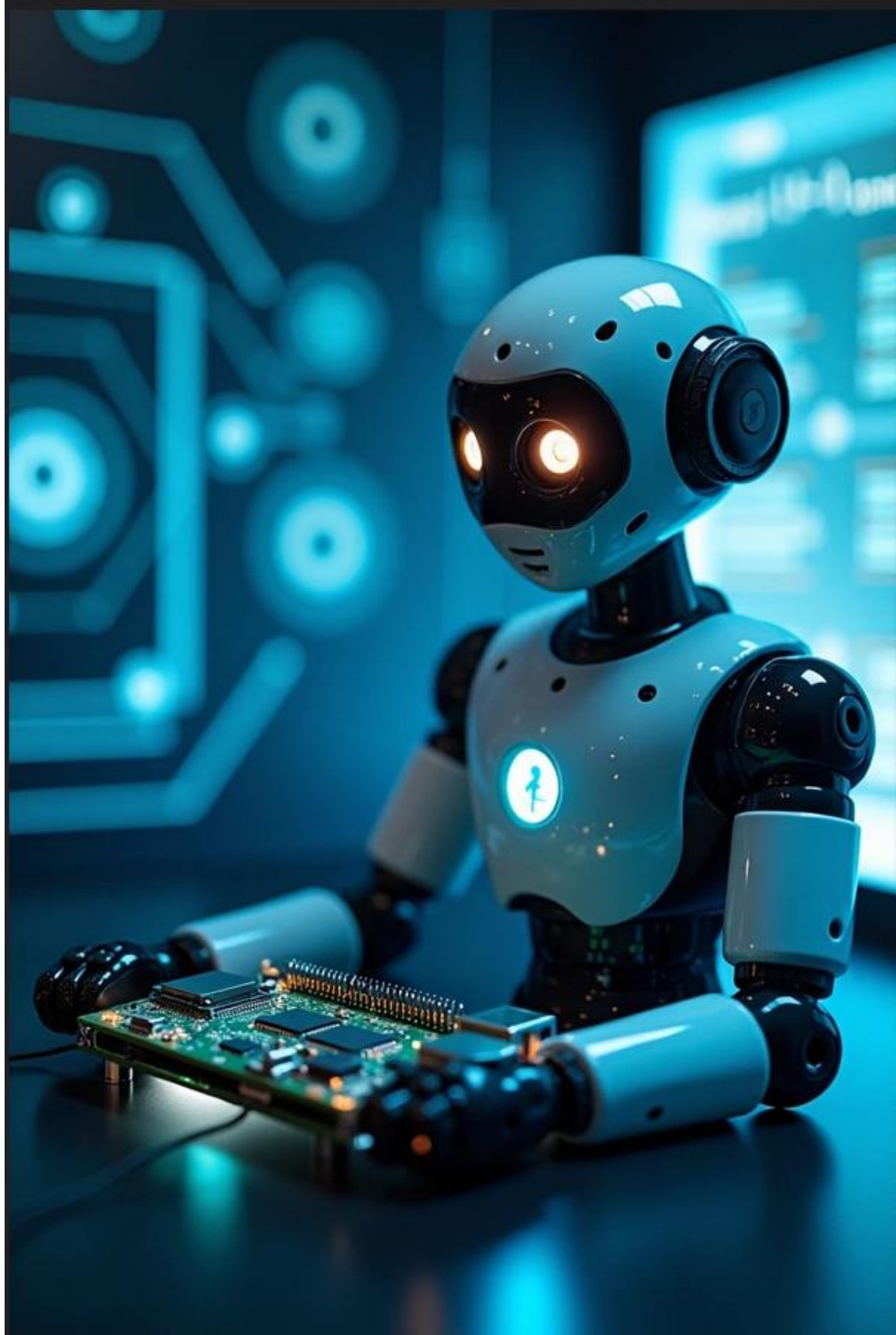
- Camera image processing
- Object, table, and human detection

•**Navigation & Autonomy**

- LiDAR data processing
- Mapping and localization (SLAM)
- Path planning and obstacle avoidance

•**Decision Making**

- Task scheduling (navigation, serving, stopping)
- Interaction logic



Power Budget & Battery Sizing

Component	Operating Voltage	Max Current	Consumed Power
2 Hoverboard	42V	1.2A	50.4W
ESP	5V	0.5A	2.5W
Raspberry pi	5V	1A	5W
RPLIDAR A1	5V	0.8A	4W
USB Camera	5V	0.5A	2.5W
2 Stepper Motors	12V	0.35A	4.2W
Total			68.6W

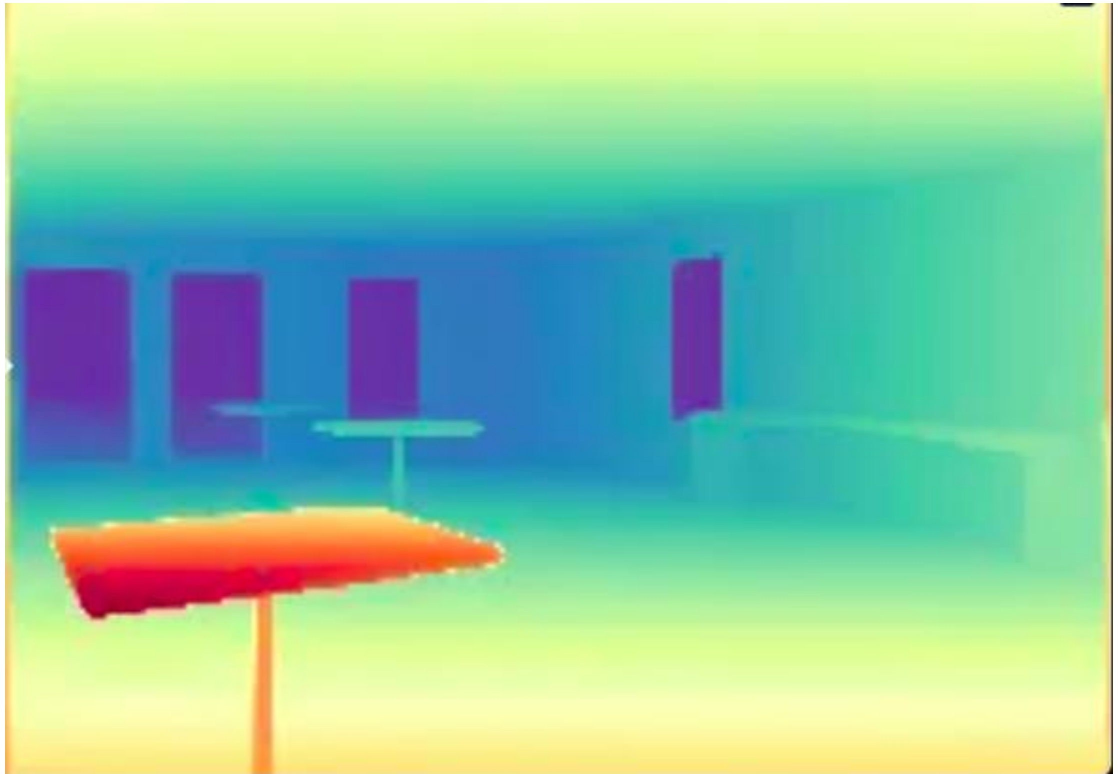
•**Nominal voltage:** 36 V (10S Li-ion)
•**Capacity: 4400 mAh (4.4 Ah)**
•**Fully charged voltage: 42 V**
•**Energy:**
E=V×Ah=36×4.4=158.4 Wh

⏱ **Expected runtime**

$$\text{Runtime} = \frac{158.4}{68.6} \approx 2.3\ hours$$

Even after losses (buck converters, motor drivers, inefficiencies ~20%):
➞ **~1.8 hours realistic runtime**

Software, Navigation & Communication

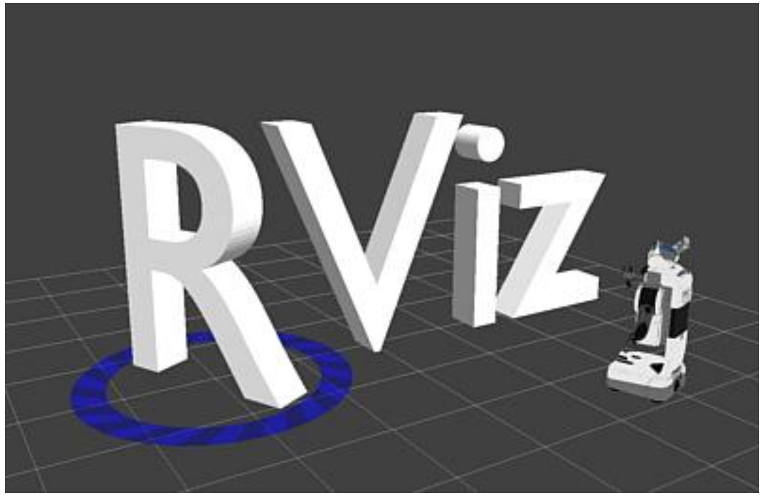


Software and Sensors communication

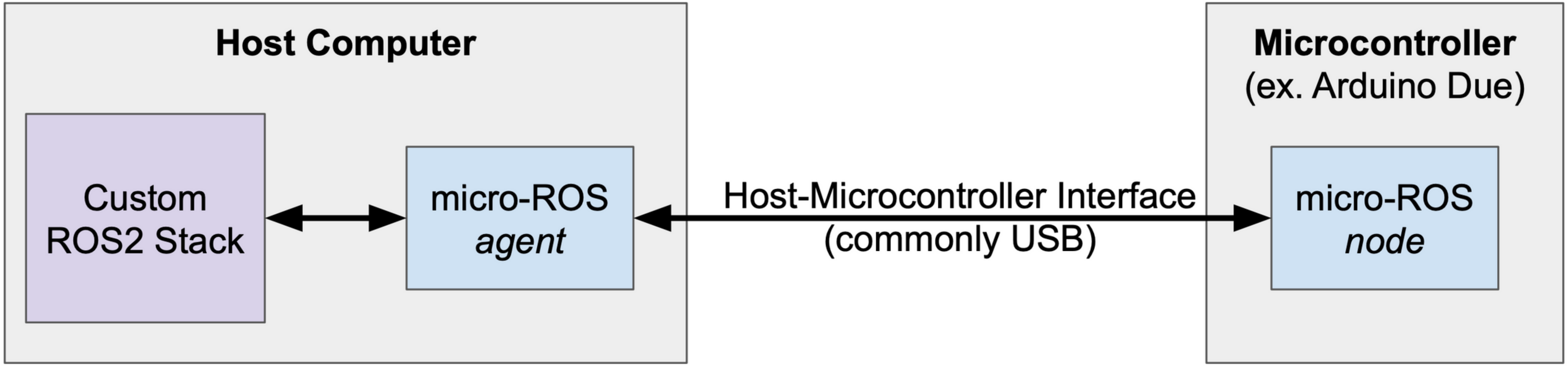
ROS2 Jazzy

ROS (Robot Operating System) is a middleware framework that enables real-time communication between robot hardware, perception, simulation ,planning, and control.
It is also widely used in autonomous navigation systems and
Large ecosystems eg. SLAM, navigation, visualization & simulation tools

Sensors (eg. IMU, Lidar and wheel encoders), planners, and controllers run as independent nodes with Real-Time Communication which allows Publish/Subscribe model that enables low-latency data flow



Micro-ROS extends ROS 2 to microcontrollers which allows communication between sensors and software and enabling them to act as full ROS nodes.



1- sensors are connected to microcontroller

2- micro-ROS node Encapsulates sensor readings into ROS2-compatible messages and publishes data via micro-ROS agent over DDS/UDP

3- micro-ROS agent Receives sensor data from microcontroller and Publishes sensor messages to the ROS2 stack

4- Robot navigation nodes (e.g., PPO, A*, obstacle avoidance) subscribe to these topics and Receives sensor data in real time

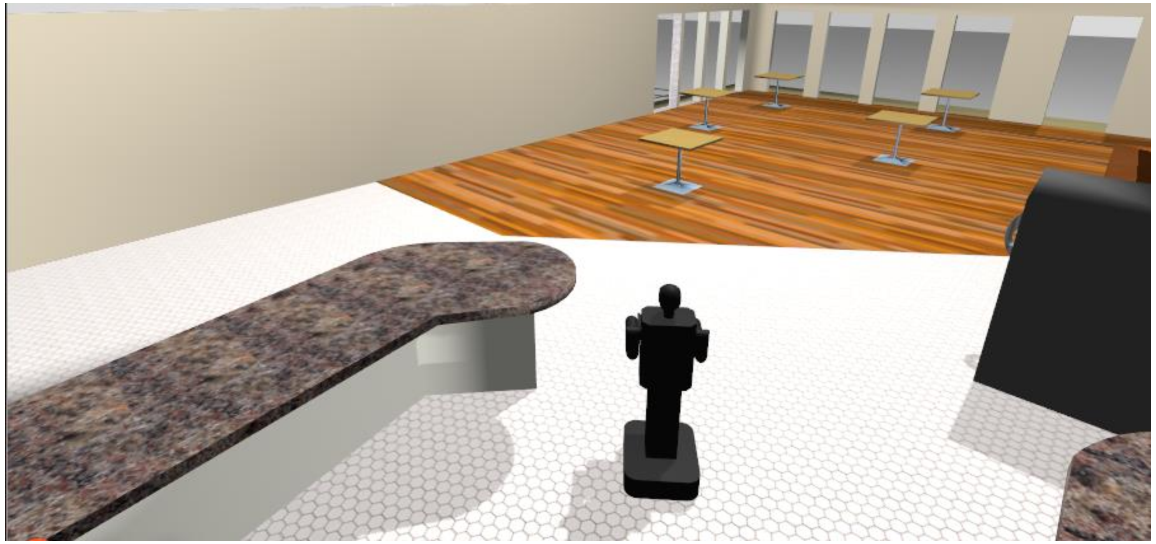
Robot Simulation

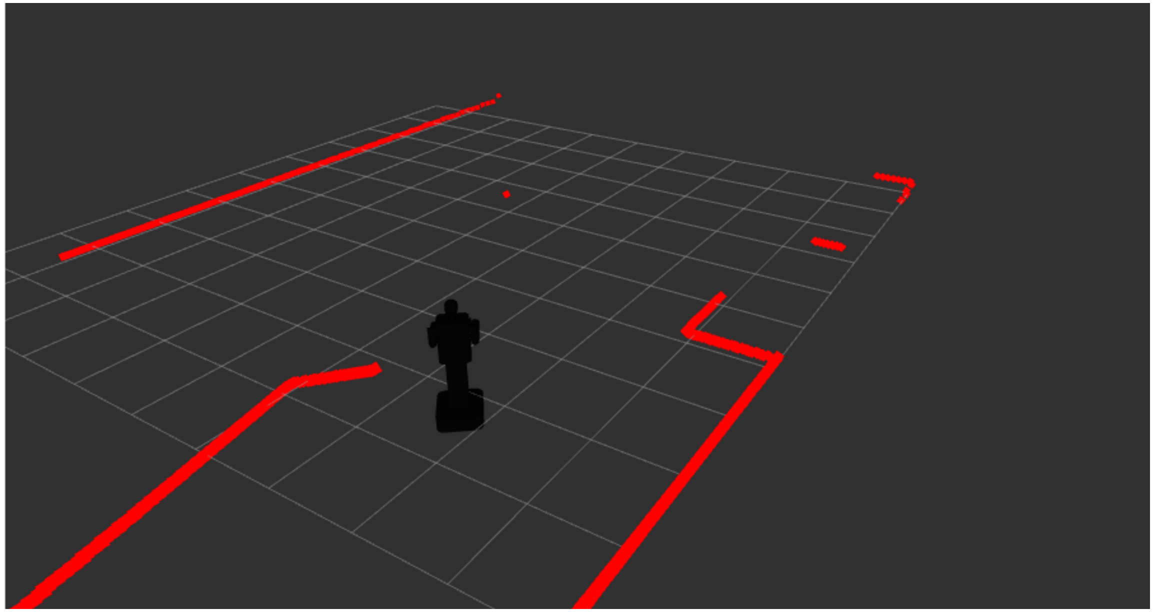
Gazebo / Rviz

Gazebo sim: Gazebo is an advanced robot simulation tool used to simulate enviroment and physics. it simulates restaurant environment: static walls, tables, and dynamic obstacles (humans) where path planning can be tested before deploying to real robot using simulated sensors with noise to simulate real world non-ideal sensors

Rviz: RViz (ROS Visualization) is a 3D visualization tool for robots and sensor data in ROS/ROS2.

Both are used to test the navigation system

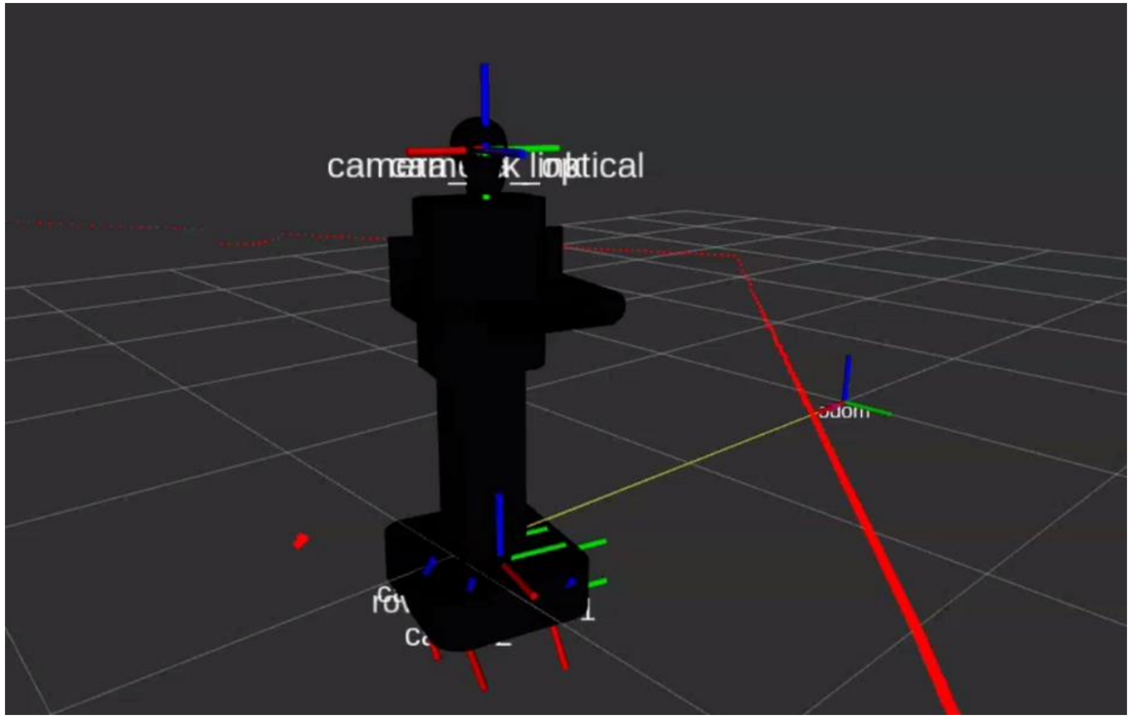




Sensors Fusion

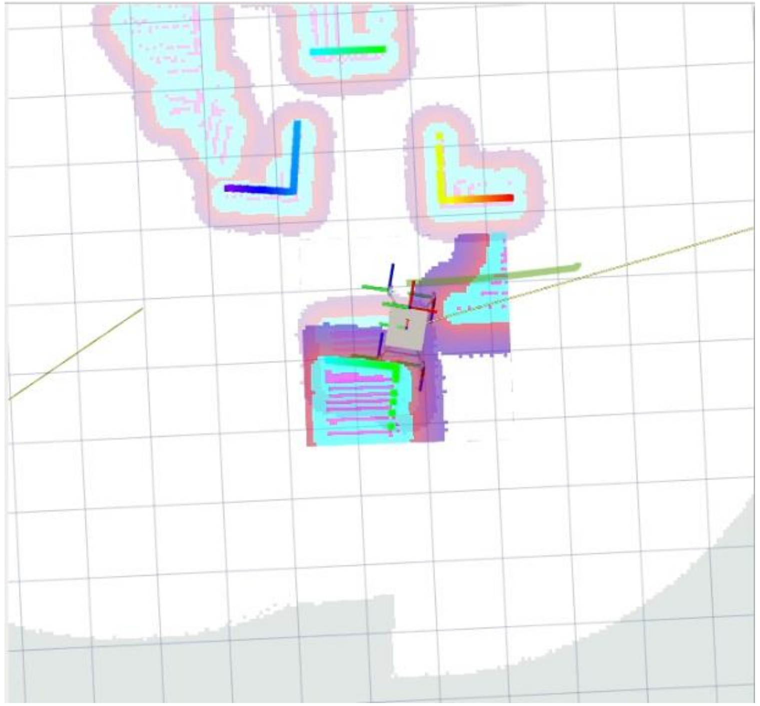
Single-ekf (extended Kalman filter)

Used to estimate robot pose (position + orientation) from noisy sensor data (IMU, wheel encoders) for non-Linear systems by fusing odometry and predicting how the robot moved since the last step and correct these measurments using IMU yaw



Navigation & obstacle avoidance

instead of depending on a searching algorithm alone, our navigation algorithm merges searching algorithms for static map along with RL for dynamic enviroment for safer navigation



A* algorithm is used to plan a path to the desired position given distance and yaw, after path is planned sucessfully, **PPO** (proximity policy iteration) RL algorithm navigates the robot through dynamic obstacles which are not part of the map

Training

PPO is trained with multiple episodes to simulate different scenarios

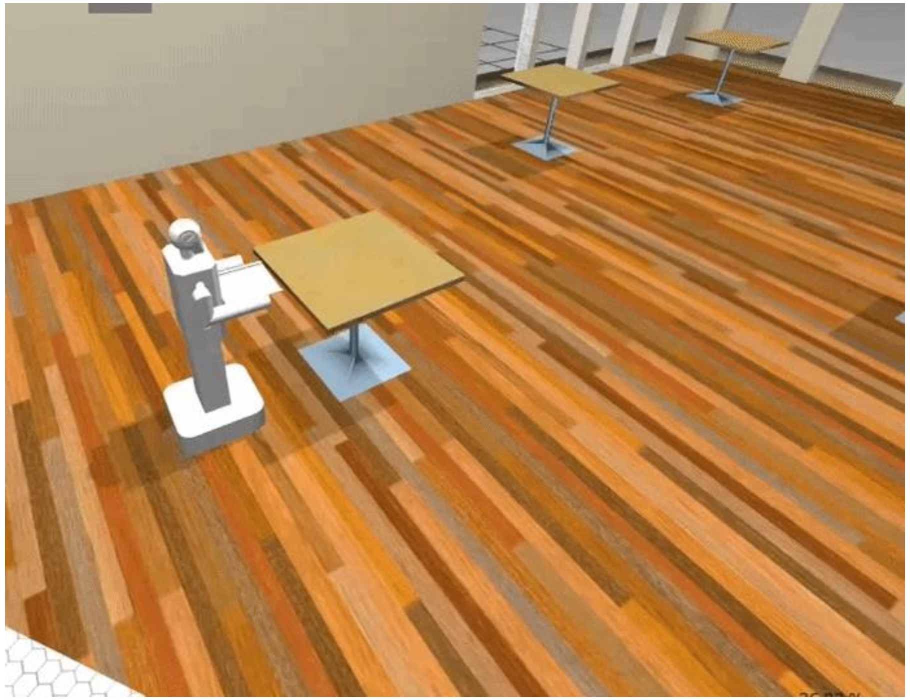
The Reward system combines: Time taken, Obstacle avoidance, Close to A* path, Close to goal, Goal reached

Weighted Reward equation:

$$R_t = w_1 \cdot R_{time} + w_2 \cdot R_{collision} + w_3 \cdot R_{path} + w_4 \cdot R_{goal\_dist} + w_5 \cdot R_{goal\_reach}$$

yelllow path is the A* path planned on static environment with gray obstacles
but while robot’s move, the navigation depends on PPO algortihm for red (dynamic) obstacle avoidance by changing it’s path or waiting until safe

Early PPO training



The robot reaching it's goal and stopping in required position according to lidar readings

The robot safely avoiding obstacle through it's way

CYBERWAIT – Smart Restaurant Ordering System

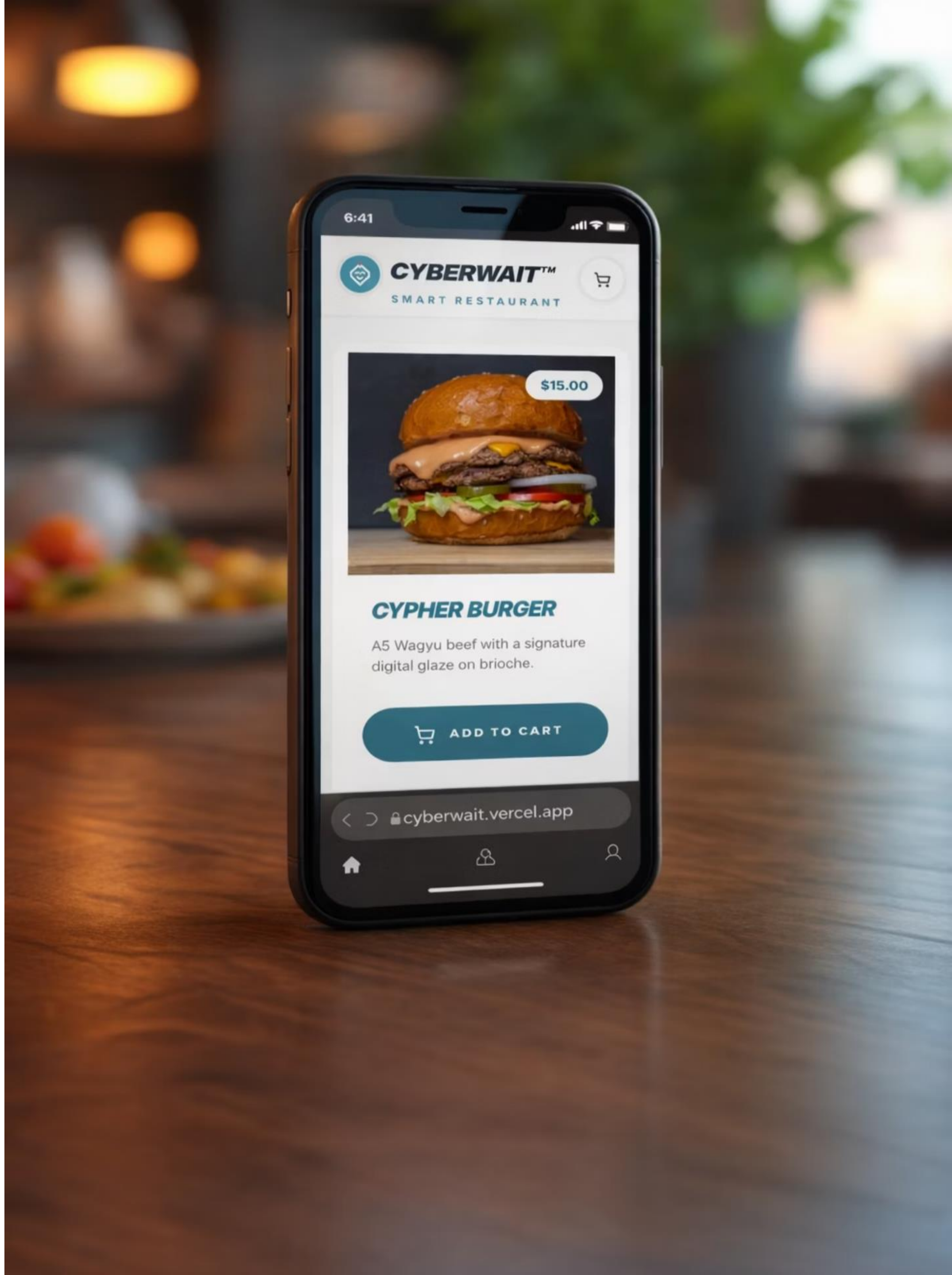
Automated digital dining experience.

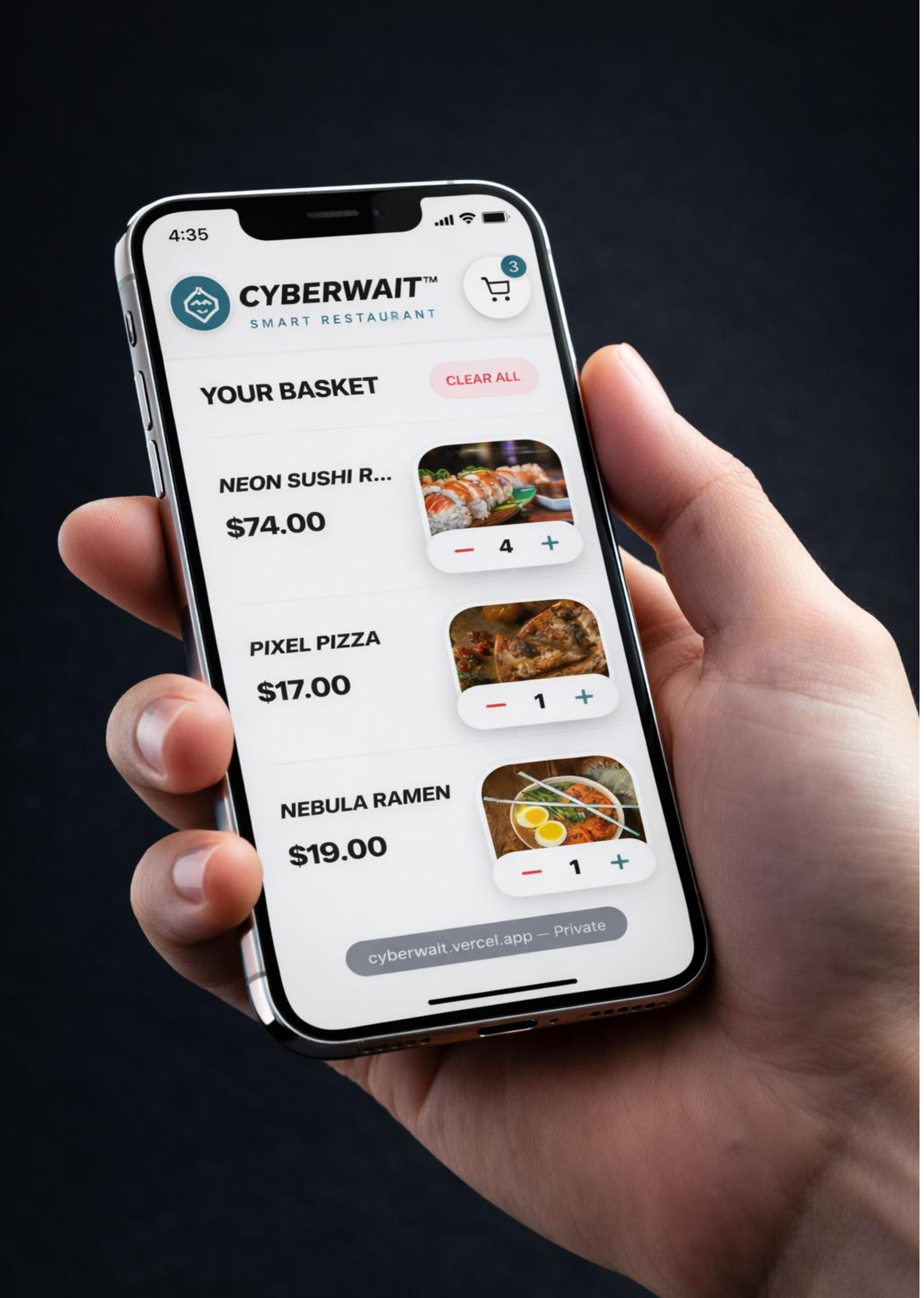
Solution Overview

CYBERWAIT provides a fully digital, end-to-end restaurant ordering and ordering and service solution. Scan the qr-code and start the experience. experience.

Interactive Menu Browsing

Interactive digital menu with rich visuals and real-time pricing.





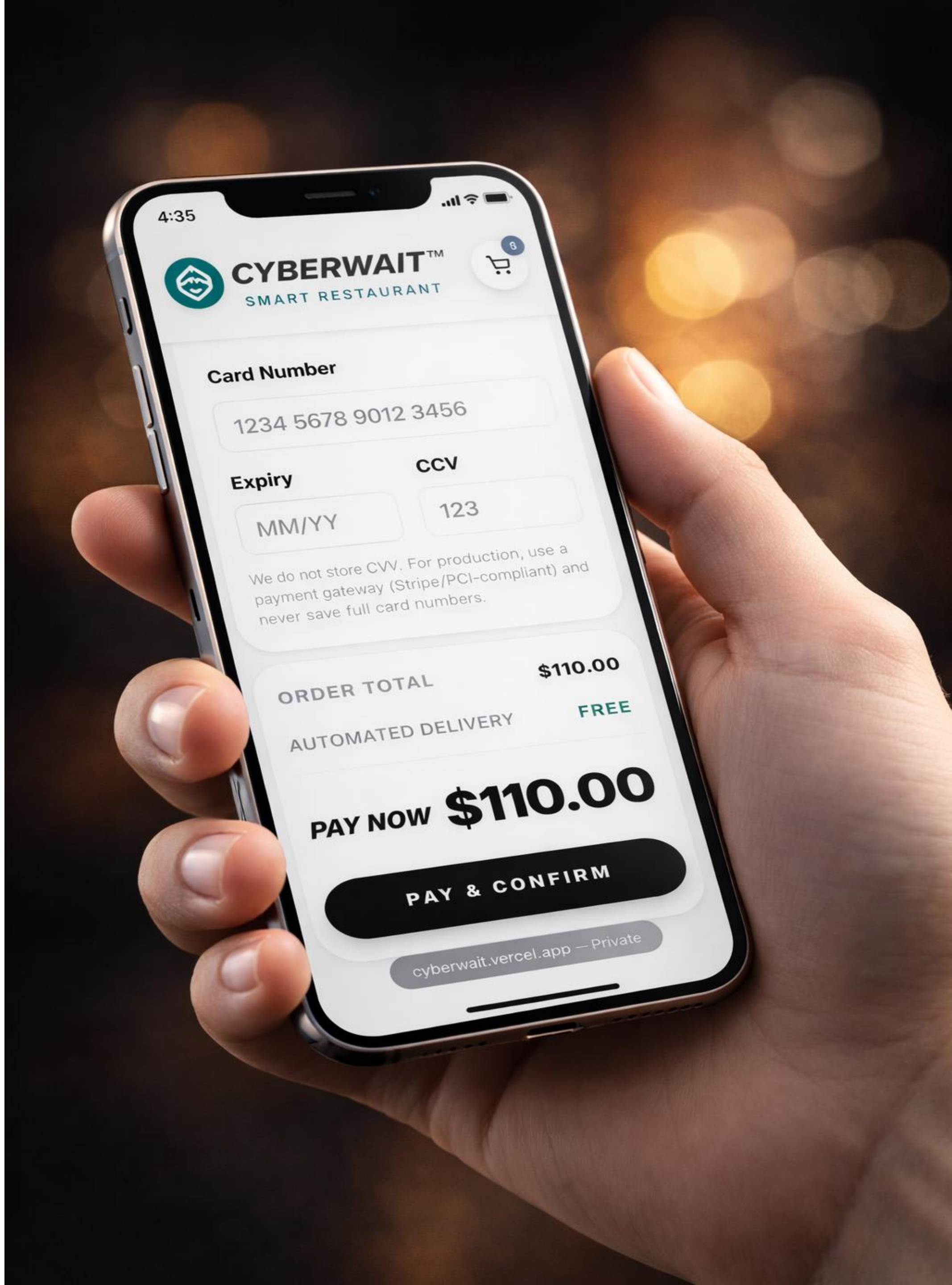
Streamlined Cart Management

Users can review, modify, and manage orders seamlessly before checkout.

checkout.

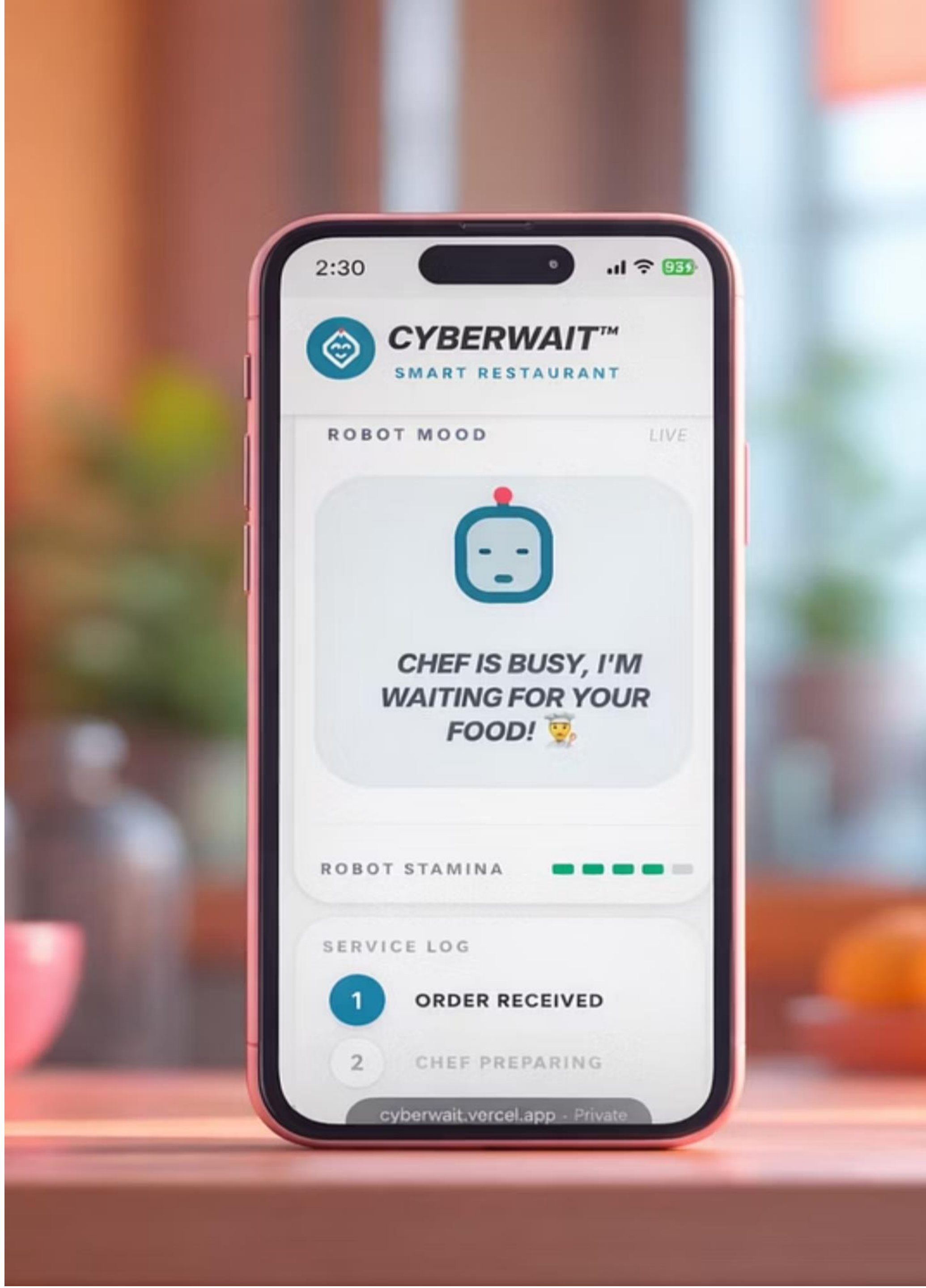
Secure & Integrated Payment

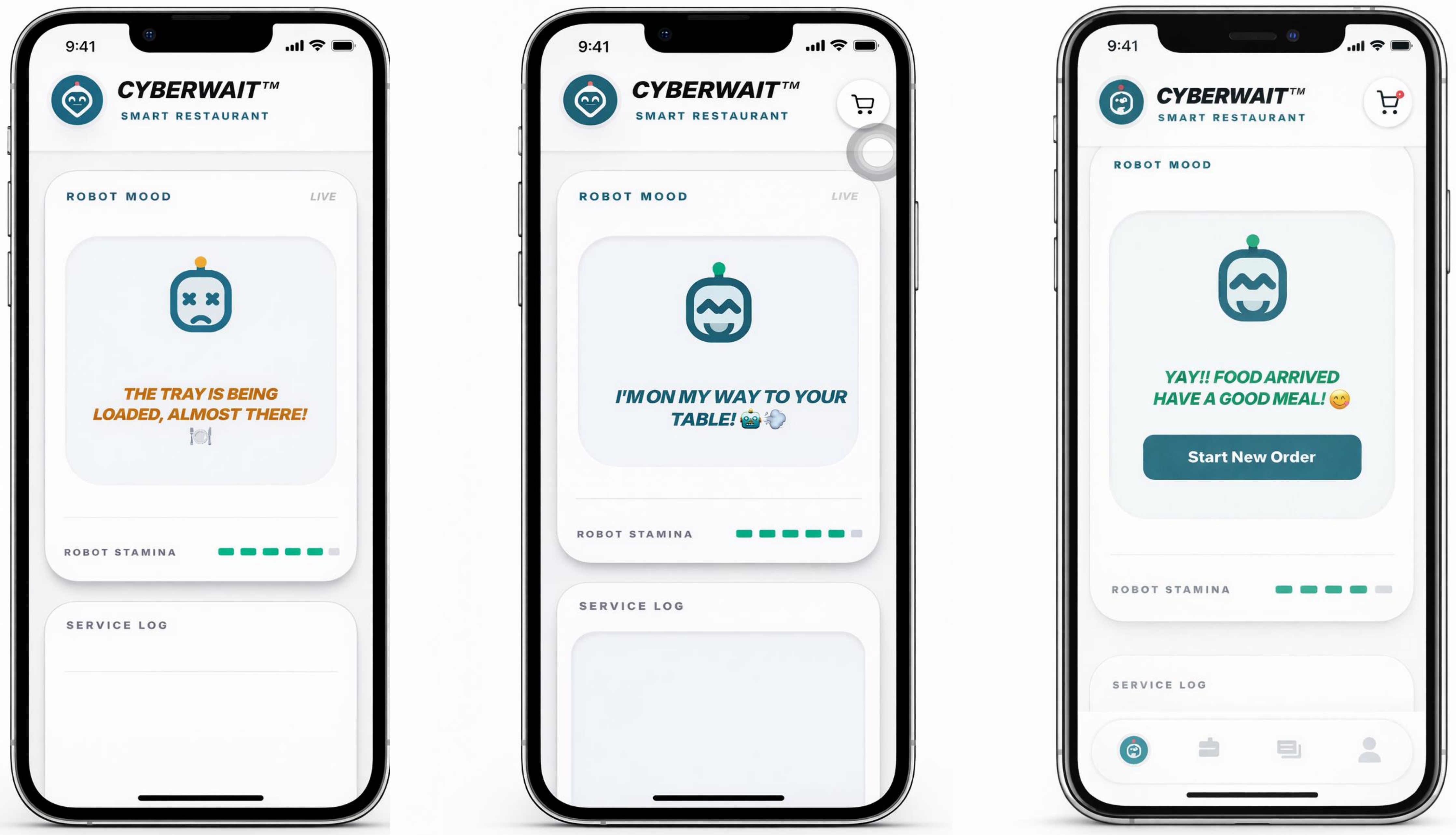
Secure and simplified payment processing integrated into the ordering flow.



Real-time Order Status & Interaction

Real-time order status with intelligent robot feedback for user engagement.



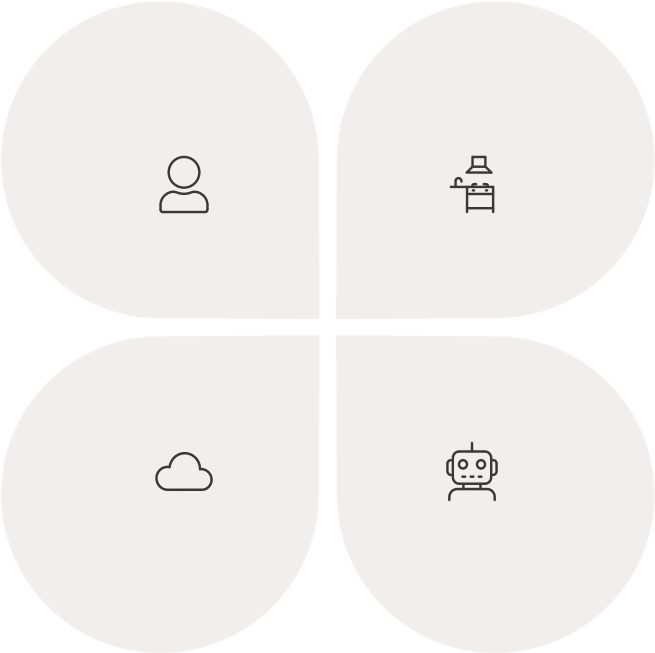

9:41
CYBERWAIT™
SMART RESTAURANT
ROBOT MOOD
LIVE
THE TRAY IS BEING LOADED, ALMOST THERE!
ROBOT STAMINA
SERVICE LOG
9:41
CYBERWAIT™
SMART RESTAURANT
ROBOT MOOD
LIVE
I'M ON MY WAY TO YOUR TABLE!
ROBOT STAMINA
SERVICE LOG
9:41
CYBERWAIT™
SMART RESTAURANT
ROBOT MOOD
YAY!! FOOD ARRIVED
HAVE A GOOD MEAL!
Start New Order
ROBOT STAMINA
SERVICE LOG

User Interface

Seamless customer interaction.

Kitchen Operations

Efficient order preparation.



Integrated Backend

Centralized data and coordination.

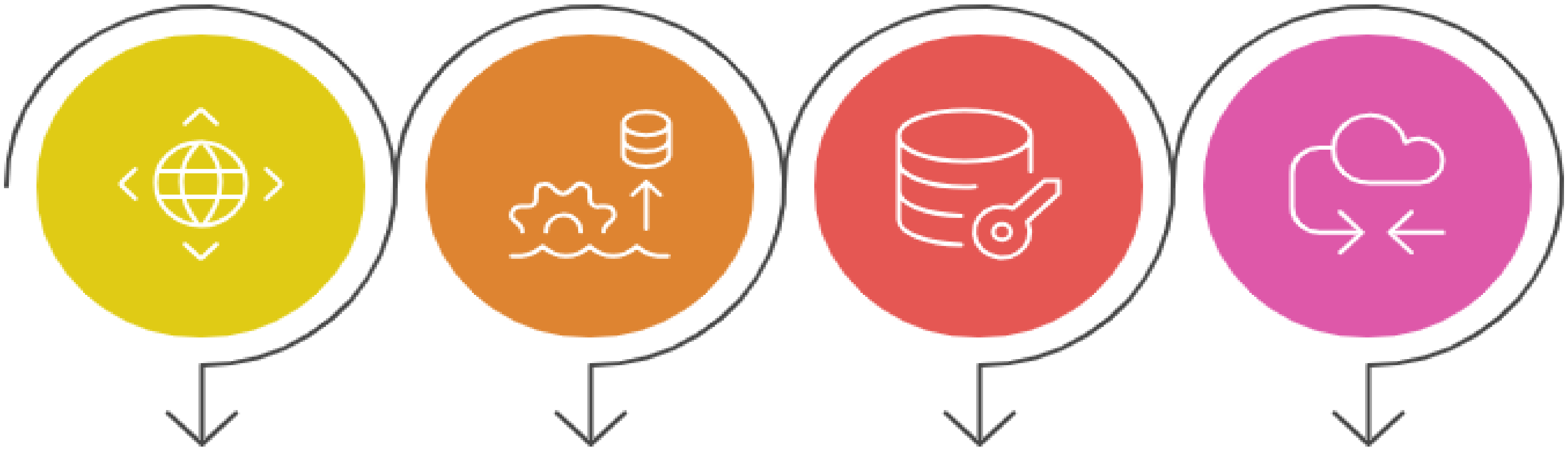
Service Robotics

Automated delivery and assistance.

Intelligent System Coordination

The CYBERWAIT™ backend intelligently orchestrates interactions between users, kitchen staff, and service robots for peak efficiency.

Project Technologies



Frontend

React.js, Tailwind CSS, and PWA for a responsive user interface.

Backend

Node.js, Express.js, and Vercel Serverless Functions for scalable backend deployment.

Database

Supabase (PostgreSQL), Row Level Security, and JWT Authentication for secure data access.

Deployment

Vercel, Git & GitHub, and Figma for hosting, version control, and UI/UX design.

Robust Backend & Data Management

Robust backend powered by a cloud database ensuring real-time synchronization between users, kitchen, and service logic.

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public.orders | public.order_items | public.menu | +

Filter Sort Insert 1 RLS policy Index Advisor Enable Realtime Role postgr

	id int4	name text	category text	price numeric	description text
	1	Sushi Roll	Meals	18.50	Fresh salmon with electric wasabi pear
	2	Beef Burger	Meals	15.00	A5 Wagyu beef with a signature digital
	3	Garden Salad	Meals	12.50	Organic greens with micro-herbs and c
	5	Espresso	Drinks	6.50	Double shot of single-origin smart brev
	8	Birthday Cake	Desserts	12.00	Layers of translucent fruit jelly and Ma
	10	French Fries	Meals	7.00	Truffle-dusted potato wedges with a m
	11	Ramen	Meals	19.00	Midnight-blue broth with glowing bam
	12	Mexican Tacos	Meals	14.50	One mild, one wild. Precision-balanced
	13	Ranch Pizza	Meals	17.00	Perfectly placed pepperonis on a perfe
	15	Coca Cola	Drinks	5.00	Zero-sugar, deep-black sparkling refre
	17	Donuts	Desserts	11.00	Circuit-board icing with popping cand

Page 1 of 1 100 rows 11 records Da

ahmedgewilli's Org FREE / ahmedgewilli's Project / main PRODUCTION Connect

Feedback Search... ⌘K

Table Editor

public.orders | public.order_items | public.menu | +

schema public

+ New table

Search tables...

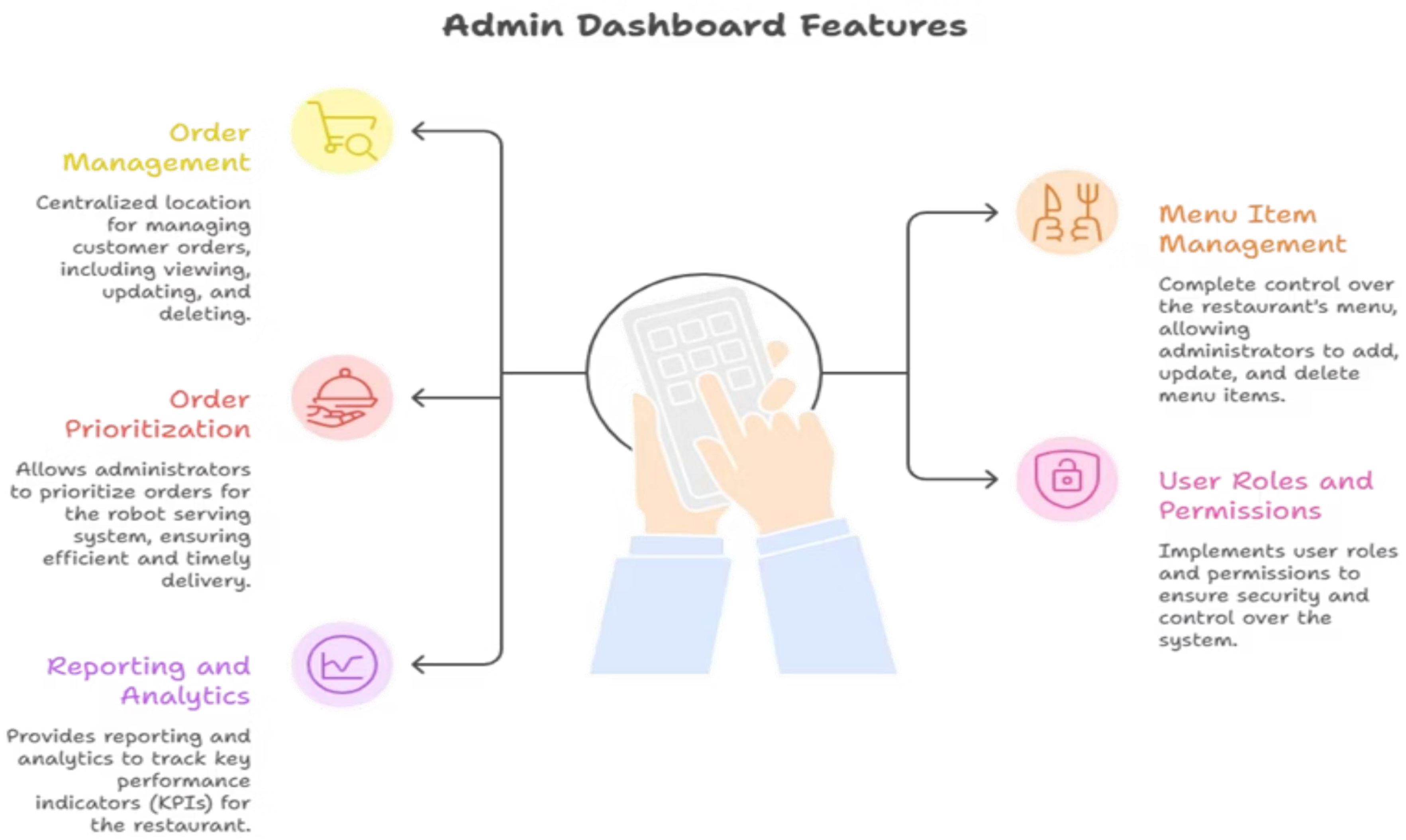
- menu
- order_items
- orders

Filter Sort Insert

2 RLS policies Index Advisor Enable Realtime Role postgres

id uuid	profile_id uuid	status text	total numeric	created_at timestamptz	up
35321c27-f501-4cdf-858c-3c15d725b739	NULL	pending	17.00	2026-01-23 14:52:54.54479+00	20
d2769984-59a6-421b-947c-375e79380f87	NULL	pending	74.00	2026-01-23 14:27:17.427564+00	20

Key improvements:



Key improvements:

Personalized Dining Experience

System Preferences

System remembers user preferences like spicy, vegetarian, and fast meals, including deadly allergies.

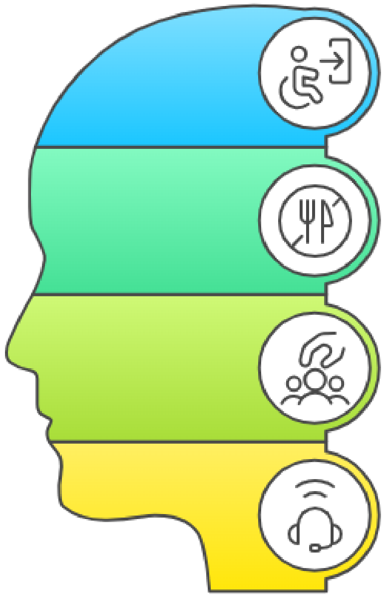
Dish Suggestions

Suggests dishes based on past orders, time of day, mood, or weather.

Powerful Benefits

Increases order value, improves user satisfaction, and brings Netflix-style personalization to dining.

Enhancing Dining with Voice Ordering



Accessibility Boost

Improves dining for elderly and visually impaired individuals

Hands-Free Dining

Allows for convenient eating without using hands

Inclusive Experience

Creates a welcoming environment for all customers

CYBERWAIT™

The system that enables voice-based ordering

The Future of Smart Dining

CYBERWAIT™ empowers restaurants by seamlessly integrating automation and intelligent systems to enhance operational efficiency, reduce service delays, and deliver a truly exceptional customer dining experience.